SAS CONTROL MODULE [STANDARD DEPLOYMENT CONTROL SYSTEM]

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Purpose

- The SAS control module controls the air bag system.

Function

Air bag system initialization diagnostics function

- When the ignition is switched ON, the SAS control module turns on the air bag/seat belt pre-tensioner system warning light in the instrument cluster for approx. **6 s** to indicate that the air bag system is performing initialization diagnostics.
- If there is no malfunction with the air bag system, the air bag/seat belt pre-tensioner system warning light turns off.

Backup power supply function

- The condenser in the SAS control module discharges even if the power supply to the SAS control module has been cut off due to a collision, and the backup power supply function operates the air bag system for a specified period of time.

Vehicle travel conditions detection function

- The SAS control module detects the signals from the following integrated sensors.
- The SAS control module outputs the detection signals from the following sensors and outputs them to the DSC HU/CM via a CAN signal.

Sensor	Signal
Low-G sensor	• Vehicle longitudinal acceleration speed signal • Vehicle lateral acceleration speed signal
Yaw rate sensor	• Vehicle steering angle speed signal

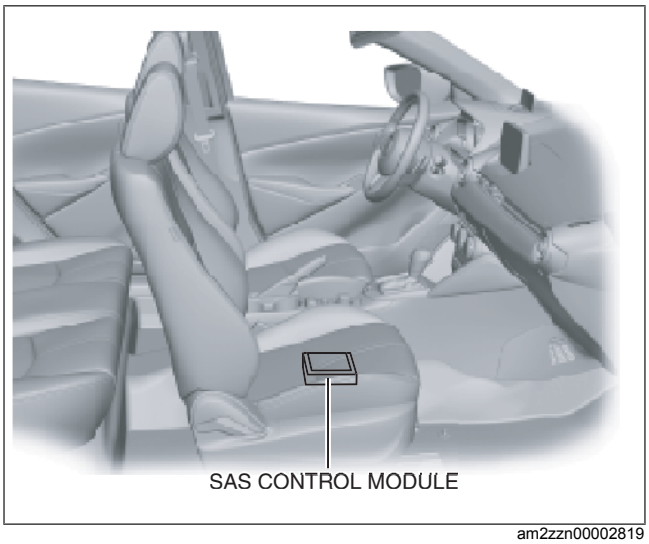
Collision detection signal output function

- The SAS control module sends the following signal to the following units when it determines that it is necessary for the air bag system to operate (deploy).
- When the signal is received, the each unit performs controls to mediate secondary injuries caused by the vehicle collision.

Related unit	Signal	Reference
• Body control module (BCM)	Secondary collision reduction signal	(See POWER DOOR LOCK SYSTEM.) (See EXTERIOR LIGHTING SYSTEM.)
• PCM • Instrument cluster • Center display	Fuel-cut request signal	(See FUEL INJECTION CONTROL SYSTEM [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See FUEL INJECTION AMOUNT CONTROL [SKYACTIV-D 1.5].) (See ENTERTAINMENT SYSTEM [WITH CENTER DISPLAY].)
• Electrical supply unit (ESU)	Collision detection unlock signal	(See POWER DOOR LOCK SYSTEM.)

Construction

- The SAS control module is located underneath the rear console.
- The following sensors are integrated with the SAS control module.
  - — Determination and safing circuit
  - — Output control circuit
- The following sensors are integrated with SAS control module.
  - — Crash sensor
  - — Yaw rate sensor
  - — Low-G sensor



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Operation

- For the SAS control module operation, refer to the AIR BAG SYSTEM. (See AIR BAG SYSTEM [STANDARD DEPLOYMENT CONTROL SYSTEM].)

Fail-safe

- If a malfunction occurs in the air bag system and DTCs are detected, the SAS control module inhibits control of the air bag system so that they do not operate (deploy) accidentally, and it flashes the air bag/seat belt pre-tensioner system warning light.